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Responding to sociotechnical controversies in education: A modest proposal toward technical democracy

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Abstract

The use of automated decision-making systems is increasing in education, applied across areas such as resourcing, enrolment, exam proctoring, and assessment. While the potential impacts of ADM are becoming widely known amongst experts such as software developers, industry leaders and policy experts, the perspectives of those impacted by ADM remain peripheral. To enrich democratic institutions and broaden expertise and participation, this paper proposes that ADM needs to be considered as a sociotechnical controversy, as part of a technical democracy approach that utilises hybrid forums. In this paper, following Callon, Lascoumes and Barthes, technical democracy refers to the process of learning through uncertainty about sociotechnical controversies, and hybrid forums refer to the specific work sites of this democratization. This paper outlines a modest, yet radical, proposal for responding to sociotechnical controversies in education, like ADM. First, it identifies key uses and concerns with ADM in education. Second, it proposes that restricted capacity for broader participation can be addressed through technical democracy. Last, it concludes by proposing that hybrid forums can create moments of democratisation through shared uncertainty, material politics, and collective experimentation that promote spaces for specialists and non-specialists, or experts and lay-people, to jointly problematise sociotechnical controversies in education.

Keywords: technical democracy, hybrid forums, education, automated decision-making, Artificial Intelligence

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Introduction

The use of automated decision-making (ADM) is becoming more widespread in education. AMD in education is applied from budget related questions to personalising the learning and assessment content of individual students (Sellar and Gulson 2021; Thompson 2021; Witzenberger and Gulson 2021). In this paper we use a very broad definition of ADM that is used in public policy areas. ADM uses 'any systems, software, or process that use computation to aid or replace government decisions, judgments, and/or policy implementation that impact opportunities, access, liberties, rights, and/or safety' (Richardson, 2021). As such, ADM uses systems that 'can involve predicting, classifying, optimizing, identifying, and/or recommending' (Richardson, 2021), and can sometimes involve forms of Artificial Intelligence (AI). We thus see ADM in education as part of sociotechnical infrastructures which delegate varied forms of decision-making to automated procedures.

During the COVID-19 pandemic, education technology (EdTech) underpinned by ADM became an important factor in the education sector, providing schooling while schools were closed. The use of ADM during the pandemic included machine learning as part of online classroom settings such as Google Classroom (Perrotta et al. 2021), and 'online examination proctoring' using biometric technologies (Selwyn et al. 2021). The COVID-19 pandemic made the use of ADM hyper visible to a diverse set of stakeholders and policy actors as these systems became embedded across organisational, classroom, and home environments. In particular, this widespread use during the pandemic made it possible for laypeople, or non-specialists, to become more aware of issues using ADM that critical scholars and commentators have seen as controversial in education (Williamson, Macgilchrist, and Potter 2021). In education,

controversies include the use of facial recognition in schools for attendance (Edvardsen, 2019), and enrolment algorithms for scholarships (Engler, 2021). Issues arising from these uses of ADM include transparency of decision-making, data privacy, and understanding proprietary and 'black-boxed' algorithms (Pasquale 2015). A black box system is one in which even experts may not fully understand how inputs become outputs' (Waldam 2019, p.618). In this paper, as automated decision making is a sociotechnical undertaking, inputs include everything from human decisions prior to coding and data selection, to the type of calculations (e.g., forms of AI such as deep learning approaches to machine learning), as well as the ongoing human labour required to maintain and operationalise these systems (Gray & Suri, 2019).

As these issues of black box systems become more apparent, they raise a key problem with the use of ADM in education. As ADM has become more prevalent in the education sector, even prior to the pandemic, there has not been a similar increase in opportunities to open these emerging technologies to joint scrutiny and understanding – especially with the people who stand to benefit, or be harmed, from their use of these systems. As such, the premise of this paper is that the proliferation of ADM in education highlights the need for greater critical scrutiny from non-specialists.

However, due to their complexity and opaqueness understanding algorithmic systems presents a significant methodological problem. We propose to address this problem by *forcing a conceptualisation of ADM as a sociotechnical controversy* that needs to be opened up as technical and political problem. This is important as declaring 'an issue is technical is to effectively remove it from the influence of public debate' while, conversely, 'to recognize its social dimension restores its chance of being discussed in political arenas' (Callon, Lascoumes, and Barthe 2009, 25). We aim to not only recast ADM as sociotechnical, but to re-emphasise the role of publics that emerge around matters of concern in black-boxed systems. We see this recasting and re-

emphasising as part of a response to the problem of 'displacement,' common in the interaction of technology and human systems. More specifically, displacement refers to the phenomena in which those who are impacted by system – in this paper impacted by automated decisions – are often denied opportunities to develop a critical understanding of systems (Thompson, Rutkowski, and Sellar 2018). This paper argues that displacement is part of the challenge for what Callon, Lascoumes and Barthe (2009) describe as technical democracy.

Technical democracy is about 'using the generative force of controversies to enrich democracy, to bring new actors into deliberation, and to prompt new exploration by challenging established knowledge' (Landström et al. 2011, 1618). Specifically, a technical democracy approach to ADM seeks to advance collective learning and experimentation about socio-technical controversies, particularly those relating to public institutions. This paper illustrates how ADM in education can be examined, understood and responded to, through the idea of technical democracy and the specific method of hybrid forums. This paper proceeds as follows. First, we locate this paper in the fields of digital policy sociology and critical data studies, and identify key uses and concerns with automated decision-making in education. We contend that there is restricted capacity for non-experts to participate in deciding how, when, where and if to use ADM. Second, we propose that this lack of capacity to participate can be addressed through technical democracy that identifies black box systems, and investigates why they are there, what they do and the consequences of using them. In this paper we focus on the potential of hybrid forums to 'organise cooperative research between specialists and concerned groups' (Callon, Lascoumes, and Barthe 2009, 250). Hybrid forums present an intriguing means to create 'moments of democratization' (Farías and Blok 2016) in sociotechnical controversies that respond to the problem of displacement. The aim is to not create new consultative mechanisms, and consensus, but rather following Rancière (2003) to make incisive interventions into matters of concern, and new modes and sites of cooperation to generate uncertainties about ADM. Drawing on work by

Farías and Blok (2016), we outline how hybrid forums can reconfigure, recast, and re-emphasise cooperation between specialists and non-specialists through enacting the following practices: shared uncertainty, material politics, collective experimentation, and fragile democratization. Third, we show the potential of technical democracy and hybrid forums as a way to respond to displacement and the opaqueness of ADM in education. We suggest that hybrid forums can be sites of problematization to change thought and unsettle common sense about both ADM and issues of opaqueness. We conclude the paper by proposing that technical democracy and hybrid forums comprise a modest, yet powerful, method for collective experimentation around sociotechnical controversies, like ADM, in education.

Critical work on algorithms in education and their problems

Despite the hyperbole, ADM in education is not about sentient robots, but rather narrow applications in which many tasks are primarily procedural, low stakes, and mundane (Selwyn, Hillman, et al. 2021). Nonetheless, the increasing use of ADM in education has raised interconnected issues regarding the status of professional expertise in automated systems, bias and fairness in systems, and the role of technology corporations in education (Kizilcec and Lee 2020; Pasquale 2019; Williamson and Eynon 2020).

There are growing concerns that ADM poses challenges to the status of professional expertise in fields such as health and education. Pasquale (2019) suggests in both these fields there is a risk that automation will not only continue but expand the reduction of complex practitioner judgement to metrics. This process in education is seen as exacerbating the risk that there will be no human oversight of high stakes decisions that impact students (Selwyn, Hillman, et al. 2021).

This notion of oversight is often connected to the issues of biased data, fairness and the role of humans in automated decision-making. Issues of bias are not so much that data are biased, for all data used in ADM such as machine learning has bias, for this is how the algorithms learn (Amoore, 2020). Rather, the main issue is that the features of databases will lead to pernicious and unfair outcomes. For example, if the data used in training facial recognition is featuring white people only, or when the outcomes of automated systems are based on 'historical patterns of discrimination and classification, which often construct harmful representations of people based on perceived differences, are reflected in the assumptions and data that inform AI systems, often resulting in allocative harms' (Whittaker et al. 2018, 25). In relation to the role of humans in ADM a key issue refers to the social, cultural and racial homogeneity of engineers developing ADM systems. Developer communities are often overwhelmingly white, male and in their late 20s, and early 30s (Campalo et al. 2017). As such, there is a causal link identified between bias and unfair ADM and human relations and biases.

These issues of fairness, and developer and data bias, are exacerbated in the use of proprietary systems that are common in education. A major component of the increased use of ADM in education comes through the role of technology companies, especially in the over-hyped world of AI in EdTech. While EdTech has long been a part of education decision-making, in ADM there are new opportunities for EdTech companies and other technology companies to expand their markets and influence on education, across sectors. This expansion was notable during the early stages of the COVID-19 pandemic when investment in EdTech grew from US$7B in 2019 to US$20.8B in 2021 (HolonIQ 2021). The use of proprietary ADM raises a need to recognise 'the support it has gathered from actors as diverse as state ministries, high-tech venture philanthropies and investment vehicles' (Williamson and Eynon 2020, 228). This has resulted in political and educational influence for technology companies including: i) that technology companies are now part of new expertise in education, as members of advisory groups in key

areas of policy such as assessment and data sharing across jurisdictions (Thompson 2021; Gulson, Sellar, and Webb 2022); and, ii) that with ADM, EdTech products can be conceived as actors in decision-making, where 'certain technologies are becoming proxy policy implementation devices – or *edtech policy machines*' (Williamson 2019, 396, original emphasis).

Proprietary and 'black-boxed' algorithms should, therefore, be understood as technical and political problems, an issue to which we return to below. It is clear is that while the use of ADM in education raises significant legal, social, and political issues, the capacity for non-specialists, or laypeople, to participate in deciding how, when, where and if to use ADM is underexplored. While there are emerging calls for approaches such as collective community control of technology (Crooks 2022), in what follows we intend to add a new methodological dimension to understanding and responding to ADM, based around notions of collective experimentation and shared uncertainty.

Learning through shared uncertainty: Technical democracy and hybrid forums in education

As the emergency conditions of the COVID-19 pandemic abate, questions regarding expertise and the democratisation of ADM become even more acute. As such, it is imperative to elicit ways to scrutinize the black box of ADM, with a particular emphasis on democratising possibilities of participation within the inquiry process. The necessity of scrutinising, not necessarily totally comprehending, a black box system can be addressed through a technical democracy approach that Callon, Lascoumes and Barthe (2009) describe as a way of responding to 'sociotechnical controversies' that are increasingly evident in technological societies. These 'controversies that unfold… are fostered not only by scientific and technical uncertainties but also by social uncertainties' (Callon, Lascoumes, and Barthe 2009, 26). These controversies need

new and more expansive methods to reinvigorate the public problematisation of these uncertainties.

> Controversies make possible the exploration of what we propose to call overflows engendered by the development of science and techniques. Overflows are inseparably technical and social, and they give rise to unexpected problems by giving prominence to unforeseen effects. All, specialists included, think they have clearly defined the parameters of the proposed solutions, reckon they have established sound knowledge and know-how, and are convinced they have clearly identified the groups concerned and their expectations. And then disconcerting events occur. (Callon, Lascoumes, and Barthe 2009, 28)

There is a danger in misunderstanding technical democracy as a deliberative or representative forum designed to pursue some form of majority decision-making. As we will argue below, the democracy proposed by technical democracy is far more radical than voting to endorse positions. The aim of technical democracy is learning through shared uncertainty: 'A gain must be produced. New knowledge must be acquired and shared, and new ways of thinking, seeing, and acting must be developed, pooled, and made available' (Callon, Lascoumes, and Barthe 2009, 33).

The premise of technical democracy is not the creation of more formal feedback mechanisms, such as consultation on regulation, for dealing with ADM controversies, but rather the creation of participatory experiments, and the organisation of new modes, and sites, of cooperation. Technical democracy is a forum for introducing uncertainty into what appears closed or settled. Sellar (2012) refers to this notion of introducing uncertainty as the introduction of a hesitation, or stutter, into the way we think about the effects of technology on social systems. It is an

attempt to open a problem up, to think of a problem through acknowledging the uncertainty of the world and the methods we have at our disposal to understand it.

However, a site for potentially misguided application when using technical democracy, concerns what is meant by 'democracy'. Callon and colleagues are proposing a radically different form of democracy than the Anglophone convention of seeing democracy as a representative decision-making or governance apparatus. A key theorist to underline this difference is Jacques Rancière for whom democracy is always the way(s) that a group is formed 'that do not coincide with the parties of the state or of society' (2003, 99-100). Furthermore, these groups emerge around 'disputes' or matters of concern such that the notion of the public, or publics, means the place 'where a dispute is conducted' (Rancière 2003, 100). Thus, democracy is about dissensus rather than consensus, and dissensus only exists when a group that has no voice or no qualification insists, or finds ways, of being heard. Democracy, therefore, only exists in those moments where publics form around matters of concern outside of the government apparatus and insist on being heard.

> The term democracy, then, does not strictly designate either a form of society or a form of government. 'Democratic society' is never anything, but an imaginary portrayal designed to support this or that principle of good government. Societies, today as yesterday, are organised by the play of oligarchies. There is, strictly speaking, no such thing as democratic government. Government is always exercised by the minority over the majority. The 'power of the people' is therefore necessarily also heterotopic to inegalitarian society and to oligarchic government. (Rancière 2003, 52)

For Rancière, much of what passes for democracy is oligarchy. Alongside this, expertise that demands to be heard can often be seen to side with oligarchical interests. We think this is a

particularly important point regarding technology given the commercial interests at the heart of decisions of how, when and where to use ADM in education.

Overall, for Rancière, democracy is wild and a little random – oligarchy attempts to limit this wild experiment though structuring the uncertainty of democracy through things such as politics, governance and the instantiation of an impoverished understanding of equality that assumes who must be saved/made more equal. One of the features of this impoverishment is the way that publicness, public spaces and subsequent possibilities for questioning, disruption, new alliances and so on are replaced by individualism that 'renders people insensitive to the public sphere' and the performance of dissensus (Rancière 2003, 74).

The above is a brief insight into the democracy that is aspired to through technical democracy. It is experimental and 'in the wild' as it focuses on public 'worldmaking' beyond that of assuming sociotechnical problems and their solutions. This also goes some way to explain why Callon, Lascoumbe and Barthes (2009) hold that consensus cannot be the aim of technical democracy as 'consensus is often a mask hiding relations of domination and exclusion. Democracy will not be increased by seeking agreement at any cost' (p.4). The challenge is to create dissensus, through necessitating new modes/sites of cooperation between 'specialists and laypersons' sparked by a particular sociotechnical controversy. A sociotechnical controversy short circuits the 'almost unbridgeable gulf' between specialists and non-specialists, for:

> relations between specialists and non-specialists usually bear the stamp of asymmetry. The former, imagining that they are faced with an ignorant or even obtuse public, take on the mission of enlightening and instructing the latter. The discussion established in hybrid forums wrong foots this model. It demonstrates that both categories of actors possess specific forms of knowledge (a capacity

> for diagnosis, an interpretation of the facts, a range of solutions) that mutually enrich each other. (Callon, Lascoumes, and Barthe 2009, 33)

Rancière gives us the foundation for understanding why the aim of technical democracy is not to convene deliberative groups whose aim is to forge consensus. Technical democracy is the opportunity to make an incisive, joint intervention in matters of concern already evident through democratising the dialogue between specialists and non-specialists who cooperate in a spirit of shared uncertainty and hope for collective experimentation.

Hybrid forums

Chilvers and Kearnes (2020) identify potential in reconfigured public spaces such as hybrid forums to make participation a relational, emergent notion in sociotechnical controversies. They argue for 'collective experimentation' as part of creating 'experimental collectives open to ongoing emergence and hybridity in terms of expert and public identities, human/ nonhuman relations, and the objects or matters of concern' (Chilvers and Kearnes 2020, 357). One key method of collective experimentation for Callon, Lascoumes and Barthe (2009) is the hybrid forum that functions as a form of 'research in the wild' (p.99). Hybrid forums are 'a new work site' for democratisation (p.36). These work sites experiment with 'forms of organisation and procedure intended to facilitate cooperation between specialists and laypersons, but also for giving visibility and audibility to emergent groups that lack official spokespersons' (p.36). The hybrid forum operates through uncertainty, with the aim to produce an 'event' that gives 'prominence to problems, identifying obstacles, and revealing strange and bizarre phenomena' that are not the exclusive remit of the expert or specialist (p.77). Callon, Lascoumes and Barthe argue that 'this cooperation between laypersons and specialists is even more inevitable and fruitful the closer we get to … domains in which knowledge to be produced, in one way or another, concerns the human person in his or her totality' (p.100).

Hybrid forums are meetings of the publics in formation around matters of concern (that express something of the sociotechnical controversy). That is, forums are not necessarily representative although they can involve a spokesperson, neither are they focus groups where issues are raised with disinterested participants. Hybrid forums 'do not call democracy into question; they demonstrate and express the need for more democracy, for a deepening of democracy. They are one of the particularly visible and urgent manifestations of the more general movement that calls for the democratisation of democracy' (Callon, Lascoumes, and Barthe 2009, 118) through providing the opportunity for the unheard to be heard around matters of concern.

While aimed at collective experimentation and learning, there can be a tendency for technical democracy and the hybrid forum to become the latest 'fashionable term [so that] everything becomes a hybrid forum' (Farías and Blok 2016, 544). On order to avoid this catch-all, Farías and Blok (2016) build on Callon et al.'s notion of the hybrid forum to specify four interrelated practices.

1. Shared uncertainty

The ethos/intent of a hybrid forum is the proactive co-production of problematics and knowledge. This co-production comprises a reconfigured dynamic between expertise and laypersons, yet this dynamic is not an equalisation of expertise. Farías contends that hybrid forums, even if they have the function of equalising expertise, involve a tension between 'the dialogical symmetrisation of expertise and a space for other forms of claim-making that are not oriented towards reaching socio-technical consensus' (Farías 2016, 551). For Farías, rather than providing a way in which levelling out of expertise becomes the focus, albeit still a part of these processes, the key is to ensure 'the very production and unfolding of uncertainty' (Farías 2016, 551). Shared uncertainty emerges because of the complexity of the controversies at hand. No

one system of thought (e.g., discipline or field) is satisfactory in grappling with these controversies, and expertise - such that it exists - needs to be placed into contact with the publics that have already emerged around sociotechnical controversies. The shared uncertainty is of the expert who realises that they can only grasp, and respond to, some of the problem, and the lay person who understands or experiences some of the problems but lacks a particular type of expertise to fully recognise other aspects of the problem. In other words, experts must be able, and willing, to be uncertain about their disciplinary tools and the lay community must embrace the uncertainty of expecting to be given solutions. Shared uncertainty is a mutual recognition of the need for problematics, and action, at the same time as recognising that the outcomes of the hybrid forum will necessarily be fragile and open to change.

Shared uncertainty is important for at least two reasons. First, as has been outlined above, in many forums that meet in the name of democracy, there is an imbalance between experts and non-experts. While this may be necessary in some instances where the science is unimpeachable (there is little to be gained by forming a hybrid forum to problematise gravity for example), sociotechnical controversies emerge at the limits of science, particularly where the non-specialist has unique knowledge or experience of the effects of those expert decisions. Callon, Lascoumes and Barthe (2009) refer to the ways in which 'overflows' occur as 'disconcerting events;' that emerge despite specialists thinking 'they have clearly defined the parameters of the proposed solutions, reckon they have established sound knowledge and know-how, and are convinced they have clearly identified the groups concerned and their expectations'. In these instances, many deliberative and consensus seeking conversations 'adopt the form of an authoritative discourse' such that 'they fail to respond to the questions of concerned citizens' (Callon, Lascoumes, and Barthe 2009, 28). Second, expertise is invariably linked to a narrowing of viewing and responding to problems. Part of the promise of shared uncertainty is an openness to problematics outside of the territory cultivated by expertise.

2. Material politics

Another hybrid forum practice is to 'bring things back into the picture' (Farías and Blok 2016, 545) in the pursuit of democratization. In other words, it is not enough to initiate dialogue about social justice, as if saying the words can invoke the reality, rather sociotechnical controversies must always be mediated by the material realities as work. In the context of urban geography, Farías and Blok suggest:

> we need to emphasize that the terms of relationship between experts and citizens is not solely the result of social struggles; it is mediated also by the materiality and technicality of various objects of dispute. Such objects are neither irrevocable matters of facts, nor simply malleable infrastructures at the service of dominant social interests. Instead, we need to recognize the recalcitrance, contingency and indeterminacy of urban materialities, and the way this shapes and conditions urban-political conflict. (Farías and Blok 2016, 555)

In a hybrid forum, the material in question may have many different forms. In the case of urban planning as outlined by Farías and Blok above, it can be the physical realities of space and the materials available at a given time to mediate that space. Material politics could concern objects such as a set of guidelines or a manifesto, it could concern the structural limits of building materials or the physical realities of living on a hillside prone to landslides. Importantly for our paper, they can concern algorithms, technical documentation, and the policy decisions made based on those algorithms and the data they rely on.

One example of material politics and ADM during the COVID19 pandemic was in the United Kingdom. In 2020, the Office of Qualifications and Examinations Regulation (Ofqual) created an exam algorithm to address the inability of a cohort of students in lockdown sitting their final matriculation exams which are necessary for entry to university. Instead of using teacher assessments of students, the exam regulator used a simple sorting algorithm and automated decision-making to predict students' A-level grades. The algorithm gave lower grades to students whose teachers predicted they would perform better than might be expected based on their school context and student performance in previous years. The algorithmically based grades were withdrawn after widespread student protests (Kelly 2021; Kippin and Cairney 2021). While not using artificial intelligence, the Ofqual algorithm had many features of automated decision making including predicting and recommending grades based on historical data. This high-profile sociotechnical controversy caused widespread outrage and mass condemnation and presented an opportunity for a participatory approach to better understand algorithms and their limits. The UK exam algorithm game (https://www.edufuturesstudio.com/uk-exam-algorithm-game) was produced by a multidisciplinary team to experiment with the social, ethical and technical implications of algorithms (Gulson et al. 2021). What is important is that way that the UK exam algorithm is modelled and presented as a game, allowing publics who have gathered about a matter of concern to better understand algorithmic decision-making and its limitations. For example, as the game progresses, assumptions built into decisions that aim to reduce error (and therefore unfairness) gradually accumulate based on subsequent decisions so that a model that aims to produce relative fairness in a procedural way produced the opposite. Material politics, and a focus on objects, reminds us that some problems are more intractable than others, that all technology including ADM has limits and that hybrid forums do not guarantee happiness, or that a concerned public can get what they want. What it does offer is a chance to explore and be heard, and to be taken seriously about a matter of concern.

3. Collective experimentation

The hybrid forum also becomes a cooperative site for specialists and non-specialists to interact with objects/materials that provoke collective experimentation practices. This is the point of research in the wild 'that is, amidst real world uncertainties and in collaboration with affected parties' (Farías and Blok 2016, 545). The argument is that the sociotechnical controversies are 'needed exactly because the socio-technical world is thought of as inherently immanent and indeterminate, leaving no space for finalized decisions but necessitating rather an ongoing process of new knowledge, voices, events and revisions' (Farías and Blok 2016, 545).

For example, Landström and colleagues (2011) created a hybrid forum around flood risk that had emerged as a matter of concern for a group of citizens around the erection of levee banks in the UK. Landström and colleagues (2011) created 'an experiment with public engagement' that aimed to 'co-produce knowledge through bringing scientists and concerned public together' (1618) around a controversy regarding how flood risk was estimated in the UK. Specialists used a computer-generated modelling program, based on concerns of laypeople, with the aim to let people get a 'feel for numbers' (Landström et al., 2011). This was done in a straightforward manner as the tool 'simply dropped an inch of rain on a wet catchment' that enabled them to model flood risk in a given locality. Local responses to experimenting with this model, particularly feedback emerging from local knowledge regarding the catchment itself improved the assumptions that the modellers were working with and informed the logic of modelling itself.

> The need to model small dams prompted further development of the new flow-routing algorithm. Written as a simple tool after the first meeting, connecting data with theory (`data-theoretic') for the group, it developed into a model. Emerging as an unexpected side-effect from preparing to model with CRUM2D, this data-theoretic model taught the

> scientists things about the catchment and raised interesting questions about model philosophy. (Landström et al. 2011, 1627)

Ultimately, the specialists decided not to use the original modelling software (CRUM2D) as it was unable to easily account for issues raised by the public. In this way, a better outcome was delivered because of the collective experimentation of the hybrid forum. These collective experiments regarding flood risk are examples of 'controversies [that] enrich democracy' (Callon, Lascoumes, and Barthe 2009, 28).

4. Fragile democratization

Finally, it is important to understand that what is produced in the hybrid forum is no more important than the process(es) themselves. The hybrid forum practices of shared uncertainty, material politics, and collective experimentation means that decision-making cannot be definitive, but rather continually "active, open, contingent, and revisable" (Callon, Lascoumes & Barthe 2009, 192). The intent of the hybrid forum is therefore not to end up with a proposed 'solution'. For example, to produce a manifesto, or to comment on legislation, or to produce new understanding or knowledge, or to organise a protest - although these may all be worthy outcomes. Instead, the intent is to produce new public spaces, or 'moments of democratization' that can only be produced through the cooperation of experts and lay people outside of the remit of siloed research and development.

> Shared uncertainty, material politics and collective experimentation all point to an important realization: namely, the realization that 'technical democracy' may indeed be misleading, since what is at stake are moments of democratization, a set of movements and displacements that are by definition incomplete. (Farías and Blok 2016, 545)

Hybrid forums create opportunities for publics to democratise sociotechnical controversies. The solutions or outcomes can never be pre-planned, which means the process may be unappealing to the minority who aspire to govern the majority and who may have much to lose. In that context, forcefully injecting moments of democratisation into those sociotechnical controversies emerging from the use of ADM in education works towards a technical democracy approach.

Hybrid forums and problematizing ADM in education

To paraphrase the problem that animates this paper - how can we find ways to open up new technologies that are increasingly being used in the education sector. In other words, how can we respond to the displacement of the non-technical specialists in ways that respect domain and practice expertise? Displacement is a problem related to the opaqueness, or technical complexity, of a tool that either collects and analyses data or makes decisions on collected data that are then applied to social institutions or in social settings. ADM and the role of algorithms are invariably 'black-boxes' with two characteristics that can have significant consequences in social systems. The first is where an algorithmic system's 'workings are mysterious; we can observe its inputs and outputs, but we cannot tell how one becomes the other' (Pasquale 2015, 3). Second, black boxing refers to what happens to the data after it is collected. Individuals are 'tracked ever more closely by firms and government, we have no clear idea of just how far much of this information can travel, how it used, or its consequences' (Pasquale 2015, 3).

The implications of these two characteristics are important when we consider how ADM is being, and is likely to be, used in education. A part of ADM is the use of artificial intelligence, including machine learning. This is used in many 'off-the-shelf' business intelligence and classroom platforms such as Google Classroom used globally by the education sector (Perrotta et al. 2021). Additionally, during the Covid pandemic, a form of machine learning called deep

learning, was used in 'online exam proctoring' which used biometrics to link observable gestures and facial expressions to suspicious behaviour and suspected cheating (Selwyn, O'Neill, et al. 2021). However, there is opaqueness about how an algorithm decides what is cheating and what is not and how the algorithm deals with false positive and false negatives (if at all). In short, reliability and precision remain murky. Given the high stakes for students, this opaqueness is concerning.

Even simpler uses of ADM do not involve forms of artificial intelligence. The UK exam controversy noted above was a form of ADM that used a simple sorting algorithm – and while it was publicly seen as AI this was somewhat of a distraction. The key issue of the black box was a different one from deep learning: while statistical experts could access the technical reports about the algorithm to understand the way in which the algorithm sorted grades, parents and students for the most part lacked the knowledge to investigate the algorithm in any other way aside from the consequences of application and the contrast between teacher allocated and algorithmically allocated grades. The exam controversy was exacerbated by the black-boxed nature caused by the technical expertise required to understand the decisions made. This is an ongoing problem in measurement systems in education 'where the limitations and cautions associated with their use are buried in technical reports while strong interpretations and public judgments abound' (Thompson, Rutkowski, and Sellar 2018, 56).

Another key part of opaqueness concerns who makes decisions, and where those decisions are made, in ADM. For example, in proprietary systems like Google Classroom there is a shift in pedagogical expertise away from practitioners. This is not only in the sense that expertise is displaced from classrooms and schools to the sites of software development. It is also that the way Google Classroom is structured takes away from the pedagogical agency of practitioners.

There is an opaqueness to this shift in pedagogical expertise that lies in the technical make up of a learning platform like Google Classroom (Perrotta et al. 2021).

This problem of black boxing, despite aims to advance layperson understandings of complex technical systems, is not easily resolved. For example, we note that there is a plethora of calls for new forms of literacy to emerge around ADM. Data literacy, machine learning literacy, even algorithmic literacy is just some of the examples (see Pangrazio and Sefton-Green 2020). Pangrazio and Sefton Green (2020) note the limitations of the literacy movement especially when it comes to broader structural issues of algorithms. In contrast, there is a growing understanding of the need to generate 'networked and contextual literacies' (Carmi et al. 2020) which move beyond techno-determinism towards a more collective approach.

Nonetheless, while literacy approaches have their role in explaining how ADM systems work, our suggestion to understand and respond to black boxes in ADM is methodologically different. Rather than focusing on opaqueness as a technical or regulatory problem, such as through algorithmic impact assessments or the use of open-source software (Moss et al. 2021), our way into this problem is through problematization. Stengers (2021) suggests that problematizations are experiments that bring about changes in thought and/or unsettle common sense:

> Showing that things are not as obvious as they seem is trying to render people unable to continue thinking them the way they did, to make it harder for them to stick to positions they had considered uncontroversial or common sense. When this happens, a transformation becomes possible. (Stengers 2021, 80)

Hybrid forums are sites of problematizing ADM to create new uncertainties about opaqueness, whether these be produced by deep learning algorithms or by lack of users' technical knowledge.

While the prevalence of ADM in education is a new phenomenon, opaqueness in data use is not (Lingard 2011; Lawn 2013). For example, there is much research on standardised testing that has shown the problems for students, teachers, parents and schools when those expected to either make or endorse data-based decisions are displaced from being able to critically engage with that data. As Thompson, Rutkowski and Sellar (2018) note: 'The lack of dialogue between technical and domain expert groups can lead to incomplete information and faulty interpretation' (p. 56). In the use of ADM the issue of proprietary systems reinforces this displacement as products are not often developed in collaboration with educators (Luckin 2017).

Hybrid forums can create new ways of addressing displacement around two interrelated questions: The first question is: why should experts expect to understand the concerns of non-specialists about sociotechnical controversies such as the use of ADM in education? This question is about unsettling the matter of concern, and challenging the idea that experts, whether they are developers, or administrators, can ever completely understand the concerns of the publics of ADM in education. It is possible to see this disjuncture by engaging in a different form of cooperation, such as the material politics of making things together like an algorithm game. Through such instances, uncertainty is produced as more people, with diverse expertise, understand both how an algorithm works, and the sociopolitical context in which algorithms are applied. As this happens, new publics emerge, located in myriad places, that identify specific consequences and broader systemic problems of using algorithms in already highly inequitable schooling systems, and the broader issues related to high stakes assessment in education.

The second question that hybrid forums accentuate is: why should experts expect other people to share the same concerns? This question speaks to the promise and challenge of hybrid forums as cooperative sites for collective experimentation. The integration of expertise in a hybrid forum depends on creating opportunities for broadening the expertise brought to bear on

sociotechnical controversies. We see this as a fruitful possibility in ADM in education. There are more calls to involve broad groups of stakeholders in decisions about designing and responding to automated decision-making. The rationale for these calls relates to issues raised around developer bias, lack of diversity in the design and building of systems, and lack of capacity of users to understand ADM (Whittaker et al. 2018). The premise of calls to include, for example, more diverse groups in development, assumes that there will be a congruence of what new views bring to ADM, that these views are an extension of the concerns of experts such as existing developers or critical scholars. Conversely, a hybrid forum will explicitly aim to produce an incongruence in a pursuit to democratise technical systems through problematization.

Conclusion: A modest proposal toward technical democracy

As we have outlined, technical democracy in general, and hybrid forums in particular, are a method for collective experimentation with sociotechnical controversies (Callon, Lascoumes & Barthes 2009). It is an approach that creates moments of democratisation, in the Rancièren radical sense, through shared uncertainty, material politics, and collective experimentation that generate sites for specialists and non-specialists to open problems up (Farías and Blok 2016). It values multiple forms of expertise and diverse cooperation around an already emergent matter of concern. While the focus of this paper was on the potential of forcing a conceptualisation of ADM as a sociotechnical controversy, our view is that any sociotechnical controversy in education could be a fertile space for a hybrid forum.

That said, there are some obvious points of hesitation before deciding on using hybrid forums. We do not see hybrid forums as a silver bullet, nor do we propose that every time someone critiques a sociotechnical instrument, they should immediately convene a hybrid forum. The first of these hesitations concerns the nature of what constitutes a public that forms around a matter

of concern. Hybrid forums require that the shared concern around which a public is formed is evident prior to the hybrid forum. This means that it cannot be a place for agenda setting and delivering pre-packaged solutions. As such, some 'community engagement' initiatives that institutions facilitate in order to consult to reinforce predetermined outcomes cannot be suitable for a hybrid forum. Relatedly, hybrid forum outcomes are not about making a 'clear-cut, definitive decisions' (Callon, Lascoumes, & Barthe, 2009) so it may be that those who convene the group or participate in it end up unsatisfied with the outcome and/or output - which is intended to be contingent, revisable, and open. Such is the 'never-completed enterprise of the *democratisation of democracy* – that is to say, of the people's control of their destiny' Callon, Lascoumes, & Barthe, 2009, 11).

A second hesitation is an obvious tension around expertise, about how experts are expected to 'plug in' to these matters of concern, about how much they are willing to experiment outside their expert territory, about how much they can participate in shared uncertainty when so much of their training is in methods assumed to be authoritative. Landström and colleagues (2011) reinforce this when they underline how important the 'willingness and ability of the participating scientists' is in 'the dissociation of science from the institutions that have become embroiled in controversy' to pursue 'an open-ended research agenda' (p. 1630). We would extend this to the social scientists themselves who may be more inclined to convene a hybrid forum whose expertise too can be bound up with particular methods, assumptions and expectations. A hybrid forum relies upon generating a very particular mode and site of cooperation: the ability of specialists to embrace shared uncertainty at the same time as it requires the non-specialist to be open in their thinking.

The potential of this embracing shared uncertainty can be seen if we return to the algorithm game noted above. In the process of building the game, and in the use of the game, the intent is

that expertise is unsettled in the process of new understandings and alliances becoming formed. In manufacturing uncertainty there is cooperative goal of exploring how automated decision-making systems 'can impact opportunities, access, liberties, rights, and/or safety' (Richardson, 2021). Ultimately, technical democracy and hybrid forums are about the process of democratisation, about groups of specialists and non-specialists cooperating around matters of concern. It is a powerful way to reorient critical work in fields looking at the interplay between emerging technologies and democracy.

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